242931: volcaniclastic siltstone, Mount Berry

(Duck Creek Dolomite, Wyloo Group, Ashburton Basin, Capricorn Orogen)

Location and sampling

Wyloo (SF 50-10), Farquhar (2253)
MGA Zone 50, 450198E 7513975N

Sampled on 2 August 2021

This sample was collected from outcrop on Mount Stuart Station, about 23 km northeast of Billeroo Bore, 8.6 km northwest of Quarry Hill, and 4.7 km east of Mount Berry.

Geological context

The unit sampled is the 2008–1799 Ma Duck Creek Dolomite of the Wyloo Group in the Ashburton Basin (Thorne and Johnson, 2020a). The sample was collected from a 15 km-long, fault-bounded ridge of dolomite that, until recently, was correlated with the 2420–2208 Ma Turee Creek Group (Martin, 2020; Martin et. al., 2022). However, certain field characteristics, specifically the dominant stromatolite Forms and the presence of a distinctive ironstone marker bed, suggest that it correlates with the Duck Creek Dolomite of the Wyloo Group (HJ Allen, written comm., 24 May 2022). The Duck Creek Dolomite is about 1 km thick in the type area, and consists mainly of thin- to very thick-bedded dolostone, locally with minor volcaniclastic beds, such as the present sample (Fig. 1). The Duck Creek Dolomite conformably overlies the Mount McGrath Formation, except around the western Wyloo Anticline, where it unconformably overlies the Fortescue and Hamersley Groups, and is conformably overlain by the Ashburton Formation or the June Hill Volcanics. The age of the entire Wyloo Group is poorly constrained between the ages of c. 2008 Ma for northwest-trending dykes of the Panhandle Dolerite that predate the Mount McGrath Formation (Müller et al., 2005), and 1799–1793 Ma for the June Hill Volcanics (Evans et al., 2003; Wingate et al., 2019; Thorne and Johnson, 2020b). This sample was collected to determine the age of deposition and to resolve the uncertainty in stratigraphic assignment.

Petrographic description

The sample is a thinly bedded, volcaniclastic siltstone (Fig. 1), consisting of about 90–92% matrix, 7–10% sericite, 1% quartz, and trace opaque and carbonate minerals. The matrix is composed of cryptocrystalline to microcrystalline clay minerals, probably kaolinite or dickite, which are almost colourless to brown in colour and exhibit very low birefringence. The colour variation is interpreted to be the result of partial overprinting by brown iron oxide minerals. Bedding is planar and regular at thin-section scale and individual beds range from 1 to 5 mm thick and most have diffuse but clear boundaries. Bedding is defined mainly by differences in the amount of overprinting brown clay or iron oxide minerals. Where this overprinting is incomplete, the texture consists of brown, more oxidized, narrow, wavy lenses up to 5 mm long within more pale clay minerals. A single 4.5 mm-wide brown layer at the edge of the thin section has a faintly lenticular texture, which primarily reflects variation in the proportion of iron oxide to clay minerals and sericite, and may reflect original depositional features, such as flattened shards. Very fine-grained sericite occurs within the clay as bedding-parallel, narrow wavy folia up to 1 mm long, and as much less well-aligned individual laths up to 0.05 mm long. Quartz forms transparent, angular grains up to 0.2 mm in size, mostly within discontinuous fine laminae associated with the brown layer described above. Opaque minerals, probably pyrite or magnetite, are equant (some are cubic) and occur mostly within the quartz-bearing laminae, where they are up to 0.03 mm in diameter, although smaller grains up to 0.01 mm are sparsely disseminated throughout the sample. Parallel veins up to 0.1 mm wide are orthogonal to bedding and composed of microcrystalline quartz, overprinted by trace fine-grained carbonate minerals. A volcanic provenance is inferred from the angular quartz grains and the lenticular, shard-like texture. The siltstone is not obviously deformed or metamorphosed.

Zircon morphology

Zircons isolated from this sample are mainly colourless and subhedral to euhedral. The crystals are up to 250 µm long, and equant to elongate, with aspect ratios up to 4:1. In cathodoluminescence (CL) images, all zircons exhibit concentric zoning, and some show sector zoning. A CL image of all zircons is shown in Figure 2.

Analytical details

This sample was analysed on 24–25 February 2022, using SHRIMP-B. Fifteen analyses of the M257 standard were obtained during the session, of which 13 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^{*}$ calibration uncertainty of 0.17% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^{*}$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Figure 1. **Outcrop image for sample 242931: volcaniclastic siltstone, Mount Berry**

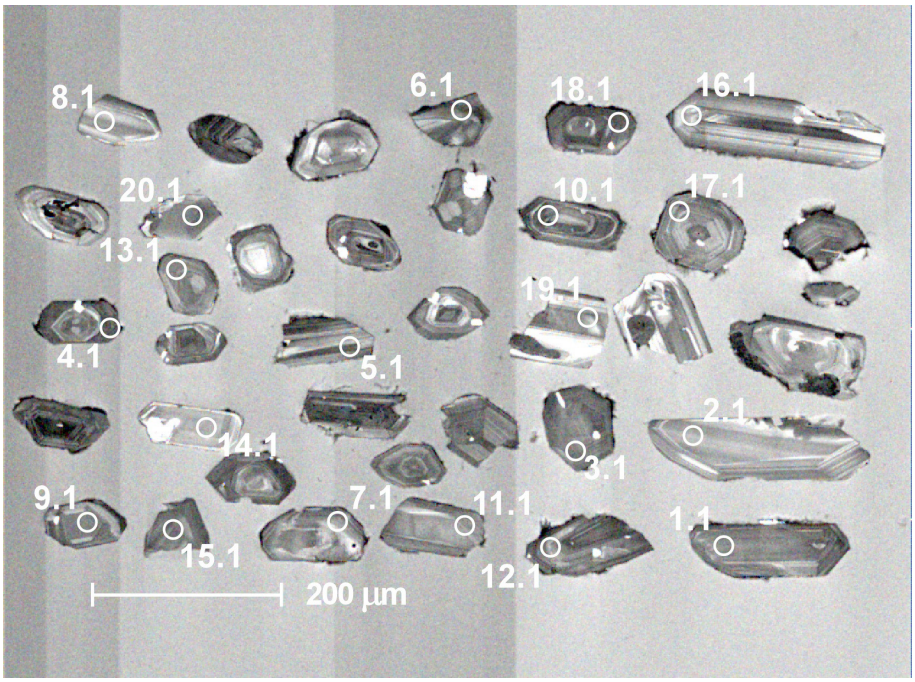

Figure 2. **Cathodoluminescence image of all zircons from sample 242931: volcaniclastic siltstone, Mount Berry. Numbered circles indicate the approximate locations of analysis sites**

Table 1. Ion microprobe analytical results for zircons from sample 242931: volcaniclastic siltstone, Mount Berry

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$	± 1σ	$^{207}Pb/^{206}Pb$	± 1σ	$^{238}U/^{206}Pb^*$	± 1σ	$^{207}Pb^*/^{206}Pb^*$	± 1σ	$^{238}U/^{206}Pb^*$ date (Ma)	±1σ	$^{207}Pb^*/^{206}Pb^*$ date (Ma)	±1σ	Disc. (%)
I	16	16.1	72	63	0.87	0.195	2.919	0.039	0.11408	0.00130	2.924	0.039	0.11238	0.00155	1896	22	1838	25	-3.1
I	19	19.1	100	143	1.43	0.059	3.013	0.034	0.11318	0.00100	3.015	0.034	0.11266	0.00106	1847	18	1843	17	-0.2
I	4	4.1	279	220	0.79	0.205	2.997	0.026	0.11457	0.00067	3.003	0.026	0.11277	0.00081	1853	14	1845	13	-0.4
I	12	12.1	157	98	0.62	0.449	2.929	0.029	0.11694	0.00089	2.943	0.029	0.11301	0.00126	1886	16	1848	20	-2.0
I	14	14.1	96	72	0.75	0.168	3.076	0.035	0.11454	0.00107	3.082	0.036	0.11307	0.00126	1812	18	1849	20	2.0
I	11	11.1	109	95	0.87	0.195	2.942	0.033	0.11484	0.00106	2.948	0.033	0.11314	0.00127	1883	19	1850	20	-1.8
I	7	7.1	97	82	0.84	0.037	3.080	0.036	0.11364	0.00114	3.081	0.036	0.11332	0.00119	1812	19	1853	19	2.2
I	2	2.1	91	122	1.34	0.316	3.005	0.065	0.11657	0.00117	3.015	0.065	0.11381	0.00153	1847	35	1861	24	0.8
I	20	20.1	157	123	0.78	-0.023	2.979	0.030	0.11420	0.00089	2.979	0.030	0.11440	0.00091	1866	16	1870	14	0.2
I	6	6.1	240	144	0.60	0.041	2.968	0.026	0.11488	0.00069	2.969	0.026	0.11452	0.00072	1871	14	1872	11	0.0
I	13	13.1	175	143	0.81	0.019	2.928	0.030	0.11509	0.00082	2.929	0.030	0.11493	0.00084	1894	17	1879	13	-0.8
I	15	15.1	194	176	0.91	0.035	3.046	0.028	0.11531	0.00079	3.047	0.028	0.11500	0.00082	1830	15	1880	13	2.7
I	10	10.1	146	166	1.14	0.255	3.145	0.032	0.11731	0.00091	3.153	0.032	0.11508	0.00113	1776	16	1881	18	5.6
I	1	1.1	137	159	1.16	0.106	3.041	0.032	0.11615	0.00097	3.044	0.032	0.11522	0.00107	1831	17	1883	17	2.8
I	17	17.1	123	113	0.92	-0.161	2.973	0.046	0.11392	0.00096	2.969	0.046	0.11533	0.00112	1872	25	1885	17	0.7
I	3	3.1	204	301	1.47	0.051	2.995	0.028	0.11594	0.00077	2.997	0.028	0.11550	0.00081	1856	15	1888	13	1.7
I	9	9.1	165	159	0.96	-0.134	2.973	0.029	0.11466	0.00083	2.969	0.029	0.11583	0.00094	1871	16	1893	15	1.1
I	8	8.1	81	118	1.46	-0.116	3.002	0.037	0.11489	0.00115	2.999	0.037	0.11590	0.00129	1855	20	1894	20	2.0
I	18	18.1	228	177	0.78	0.000	2.991	0.027	0.11620	0.00074	2.991	0.027	0.11620	0.00074	1859	15	1899	11	2.1
D	5	5.1	162	177	1.09	0.482	3.477	0.034	0.11870	0.00096	3.494	0.034	0.11449	0.00137	1622	14	1872	22	13.3

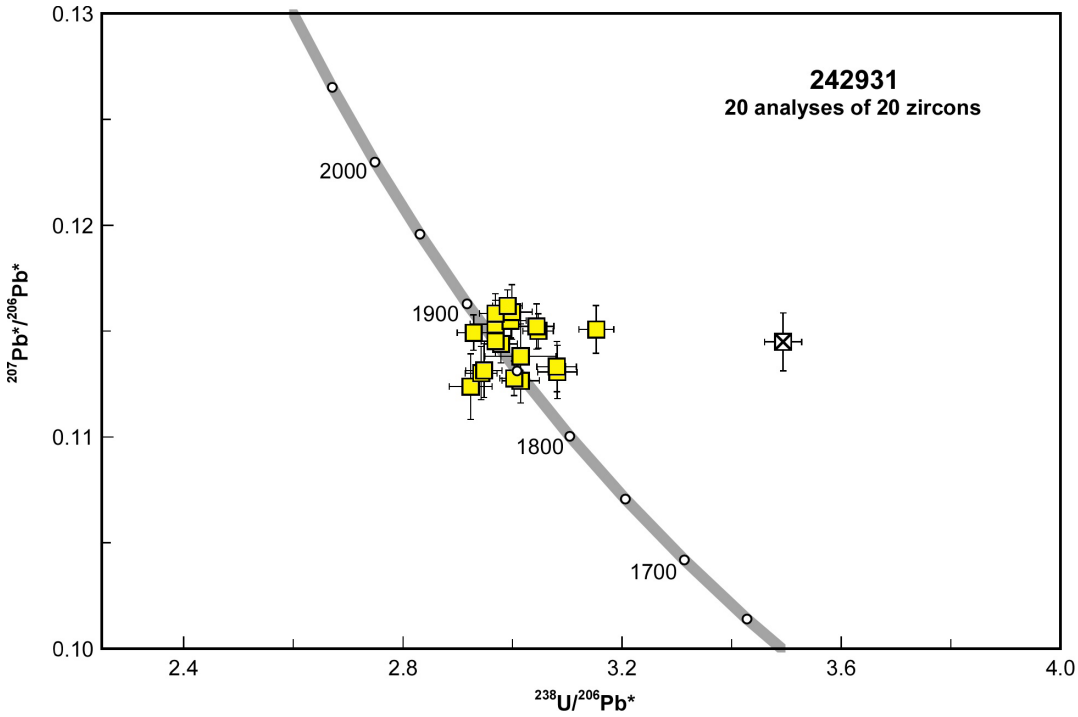

Figure 3. U–Pb analytical data for zircons from sample 242931: volcaniclastic siltstone, Mount Berry. Yellow squares indicate Group I (magmatic zircons); crossed square indicates Group D (discordance >6%)

Results

Twenty analyses were obtained from 20 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to moderately discordant (Fig. 3). One analysis is >6% discordant. The date obtained from this analysis (Group D; Table 1) is unreliable, and considered not to be geologically significant. The remaining 19 analyses form a single group, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises 19 analyses (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 1873 ± 9 Ma (MSWD = 1.5).

The date of 1873 ± 9 Ma for the 19 analyses in Group I is interpreted as the igneous crystallization age of the dominant volcanic component.

References

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Recommended reference for this publication

Wingate, MTD, Fielding, IOH, Lu, Y, Martin, DMcB and Allen, HJ 2022, 242931: volcaniclastic siltstone, Mount Berry; Geochronology Record 1898: Geological Survey of Western Australia, 4p.

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